

Assignment 5 - CP363

Moosa Ahmed, Muhammad Saim Bin Asif

169046796, 169075698

Group # 26

Equipment Test Center

Dr. Haytham Qushtom

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Verify Functional Dependencies

1. Equipment Table

- Primary Key: Equipment_ID
- Functional Dependencies:
 - Equipment_ID → Name, Model_Number, Manufacturer, Status

2. Client Table

- Primary Key: Client_ID
- Functional Dependencies:
 - Client_ID → First_Name, Last_Name, Contact_Email, Phone_Number, Organization
 - Contact_Email → Client_ID, First_Name, Last_Name, Phone_Number, Organization
 - Phone_Number → Client_ID, First_Name, Last_Name, Contact_Email, Organization

3. Technician Table

- Primary Key: Technician_ID
- Functional Dependencies:
 - Technician_ID → First_Name, Last_Name, Certification, Contact_Email, Phone_Number
 - Contact_Email → Technician_ID, First_Name, Last_Name, Certification, Phone_Number
 - Phone_Number → Technician_ID, First_Name, Last_Name, Certification, Contact_Email

4. Test Table

- Primary Key: Test_ID
- Functional Dependencies:
 - Test_ID → Test_Name, Description

5. Lab Table

- Primary Key: Lab_ID
- Functional Dependencies:
 - Lab_ID → Lab_Name, Location, Capacity

6. Booking Table

- Primary Key: Booking_ID
- Functional Dependencies:
 - Booking_ID → Booking_Date, Client_ID, Lab_ID, Start_Time, End_Time, Status

7. Test_Session Table

- Primary Key: Session_ID
- Functional Dependencies:
 - Session_ID → Test_Date, Result, Duration, Technician_ID, Test_ID

8. Report Table

- Primary Key: Report_ID
- Functional Dependencies:
 - Report_ID → Report_Date, Findings, Approved_By, Session_ID
 - Session_ID → Report_ID, Report_Date, Findings, Approved_By

9. Invoice Table

- Primary Key: Invoice_ID
- Functional Dependencies:
 - Invoice_ID → Booking_ID, Invoice_Date, Amount, Payment_Status
 - Booking_ID → Invoice_ID, Invoice_Date, Amount, Payment_Status

10. Maintenance_Record Table

- Primary Key: Maintenance_ID
- Functional Dependencies:
 - Maintenance_ID → Equipment_ID, Maintenance_Date, Technician_ID, Notes

11. Booking_Equipment Table

- Primary Key: (Booking_ID, Equipment_ID)
- Functional Dependencies:
 - (Booking_ID, Equipment_ID) → (no additional attributes)

12. TestSession_Equipment Table

- Primary Key: (Session_ID, Equipment_ID)
- Functional Dependencies:
 - (Session_ID, Equipment_ID) → (no additional attributes)

Check for 3NF compliance

Requirements:

- The table is in 2NF
- No non-prime attribute is transitively dependent on the primary key

Tables - Verifying 3NF:

```
# Equipment table
CREATE TABLE IF NOT EXISTS Equipment (
    Equipment_ID INT AUTO_INCREMENT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Model_Number VARCHAR(50) NOT NULL,
    Manufacturer VARCHAR(100) NOT NULL,
    Status VARCHAR(20) NOT NULL DEFAULT 'Available'
    CHECK (Status IN ('Available','In Use','Under Maintenance'))
);
```

Functional dependencies:

- Equipment_ID → Name, Model_Number, Manufacturer, Status

```
# Client table
CREATE TABLE IF NOT EXISTS Client (
    Client_ID INT AUTO_INCREMENT PRIMARY KEY,
    First_Name VARCHAR(50) NOT NULL,
    Last_Name VARCHAR(50) NOT NULL,
    Contact_Email VARCHAR(100) NOT NULL UNIQUE,
    Phone_Number VARCHAR(20) NOT NULL UNIQUE,
    Organization VARCHAR(100) NOT NULL
);
```

Functional dependencies:

- Client_ID → First_Name, Last_Name, Contact_Email, Phone_Number, Organization
- Contact_Email → Client_ID, First_Name, Last_Name, Phone_Number, Organization (Candidate Key)
- Phone_Number → Client_ID, First_Name, Last_Name, Contact_Email, Organization (Candidate Key)

```

# Technician table
• CREATE TABLE IF NOT EXISTS Technician (
    Technician_ID INT AUTO_INCREMENT PRIMARY KEY,
    First_Name VARCHAR(50) NOT NULL,
    Last_Name VARCHAR(50) NOT NULL,
    Certification VARCHAR(100) NOT NULL,
    Contact_Email VARCHAR(100) NOT NULL UNIQUE,
    Phone_Number VARCHAR(20) NOT NULL UNIQUE
);

```

Functional dependencies:

- Technician_ID → First_Name, Last_Name, Certification, Contact_Email, Phone_Number
- Contact_Email → Technician_ID, First_Name, Last_Name, Certification, Phone_Number (Candidate Key)
- Phone_Number → Technician_ID, First_Name, Last_Name, Certification, Contact_Email (Candidate Key)

```

# Test table
• CREATE TABLE IF NOT EXISTS Test (
    Test_ID INT AUTO_INCREMENT PRIMARY KEY,
    Test_Name VARCHAR(100) NOT NULL,
    Description VARCHAR(255) NOT NULL
);

```

Functional dependencies:

- Test_ID → Test_Name, Description

```

# Lab table
• CREATE TABLE IF NOT EXISTS Lab (
    Lab_ID INT AUTO_INCREMENT PRIMARY KEY,
    Lab_Name VARCHAR(100) NOT NULL,
    Location VARCHAR(100) NOT NULL,
    Capacity INT NOT NULL CHECK (Capacity > 0)
);

```

Functional dependencies:

- Lab_ID → Lab_Name, Location, Capacity

```

# Booking table (Client books Lab)
• CREATE TABLE IF NOT EXISTS Booking (
    Booking_ID INT AUTO_INCREMENT PRIMARY KEY,
    Booking_Date DATE NOT NULL,
    Client_ID INT NOT NULL,
    Lab_ID INT NOT NULL,
    Start_Time TIME NOT NULL,
    End_Time TIME NOT NULL,
    Status VARCHAR(20) NOT NULL DEFAULT 'Pending'
    CHECK (Status IN ('Pending', 'Confirmed', 'Completed', 'Cancelled')),
    FOREIGN KEY (Client_ID) REFERENCES Client(Client_ID),
    FOREIGN KEY (Lab_ID) REFERENCES Lab(Lab_ID)
);

```

Functional dependencies:

- Booking_ID → Booking_Date, Client_ID, Lab_ID, Start_Time, End_Time, Status

```

# Test_Session table (A session of a test conducted by a technician)
• CREATE TABLE IF NOT EXISTS Test_Session (
    Session_ID INT AUTO_INCREMENT PRIMARY KEY,
    Test_Date DATE NOT NULL,
    Result VARCHAR(100) NOT NULL,
    Duration TIME NOT NULL,
    Technician_ID INT NOT NULL,
    Test_ID INT NOT NULL,
    FOREIGN KEY (Technician_ID) REFERENCES Technician(Technician_ID),
    FOREIGN KEY (Test_ID) REFERENCES Test(Test_ID)
);

```

Functional dependencies:

- Session_ID → Test_Date, Result, Duration, Technician_ID, Test_ID

```

# Report table (Test session results approved by a technician)
• CREATE TABLE IF NOT EXISTS Report (
    Report_ID INT AUTO_INCREMENT PRIMARY KEY,
    Report_Date DATE NOT NULL,
    Findings VARCHAR(255) NOT NULL,
    Approved_By INT NOT NULL,
    Session_ID INT NOT NULL UNIQUE,
    FOREIGN KEY (Approved_By) REFERENCES Technician(Technician_ID),
    FOREIGN KEY (Session_ID) REFERENCES Test_Session(Session_ID)
);

```

Functional dependencies:

- Report_ID → Report_Date, Findings, Approved_By, Session_ID
- Session_ID → Report_ID, Report_Date, Findings, Approved_By (Candidate Key)

```

# Invoice table - FIXED: Separate primary key for auto_increment
• CREATE TABLE IF NOT EXISTS Invoice (
    Invoice_ID INT AUTO_INCREMENT PRIMARY KEY, -- New single-column primary key
    Booking_ID INT NOT NULL,
    Invoice_Date DATE NOT NULL,
    Amount DECIMAL(10,2) NOT NULL CHECK (Amount >= 0),
    Payment_Status ENUM('Paid', 'Unpaid') NOT NULL,
    FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID)
);

```

Functional dependencies:

- Invoice_ID → Booking_ID, Invoice_Date, Amount, Payment_Status
- Booking_ID → Invoice_ID, Invoice_Date, Amount, Payment_Status (Candidate Key - 1:1 relationship)

```

# Maintenance_Record table - FIXED: Separate primary key for auto_increment
• CREATE TABLE IF NOT EXISTS Maintenance_Record (
    Maintenance_ID INT AUTO_INCREMENT PRIMARY KEY, -- New single-column primary key
    Equipment_ID INT NOT NULL,
    Maintenance_Date DATE NOT NULL,
    Technician_ID INT NOT NULL,
    Notes VARCHAR(255) NOT NULL,
    FOREIGN KEY (Equipment_ID) REFERENCES Equipment(Equipment_ID),
    FOREIGN KEY (Technician_ID) REFERENCES Technician(Technician_ID)
);

```

Functional dependencies:

- Maintenance_ID → Equipment_ID, Maintenance_Date, Technician_ID, Notes

```

# Booking_Equipment junction table (many-to-many relationship)
CREATE TABLE IF NOT EXISTS Booking_Equipment (
    Booking_ID INT NOT NULL,
    Equipment_ID INT NOT NULL,
    PRIMARY KEY (Booking_ID, Equipment_ID),
    FOREIGN KEY (Booking_ID) REFERENCES Booking(Booking_ID),
    FOREIGN KEY (Equipment_ID) REFERENCES Equipment(Equipment_ID)
);

```

Functional dependencies:

- (Booking_ID, Equipment_ID) → (no additional attributes)

```

# TestSession_Equipment junction table (many-to-many relationship)
CREATE TABLE IF NOT EXISTS TestSession_Equipment (
    Session_ID INT NOT NULL,
    Equipment_ID INT NOT NULL,
    PRIMARY KEY (Session_ID, Equipment_ID),
    FOREIGN KEY (Session_ID) REFERENCES Test_Session(Session_ID),
    FOREIGN KEY (Equipment_ID) REFERENCES Equipment(Equipment_ID)
);

```

Functional dependencies:

- (Session_ID, Equipment_ID) → (no additional attributes)

Analysis:

All tables satisfy 3NF because:

- Each table has a proper primary key
- All non-key attributes are fully functionally dependent on the primary key
- There are no transitive dependencies where a non-key attribute depends on another non-key attribute
- All many-to-many relationship tables (junction tables) contain only foreign keys that together form the primary key

Table Name	Primary Key	2NF Compliant?	No Transitive Dependencies?	3NF Status
Equipment	Equipment_ID	Yes	Yes	COMPLIANT
Client	Client_ID	Yes	Yes	COMPLIANT
Technician	Technician_ID	Yes	Yes	COMPLIANT
Test	Test_ID	Yes	Yes	COMPLIANT
Lab	Lab_ID	Yes	Yes	COMPLIANT
Booking	Booking_ID	Yes	Yes	COMPLIANT
Test_Session	Session_ID	Yes	Yes	COMPLIANT
Report	Report_ID	Yes	Yes	COMPLIANT
Invoice	Invoice_ID	Yes	Yes	COMPLIANT
Maintenance_Record	Maintenance_ID	Yes	Yes	COMPLIANT
Booking_Equipment	(Booking_ID, Equipment_ID)	Yes	Yes	COMPLIANT
TestSession_Equipment	(Session_ID, Equipment_ID)	Yes	Yes	COMPLIANT

Check for BCNF compliance using our SQL tables above:

BCNF Requires: For every functional dependency $X \rightarrow Y$, X must be a superkey

1. Equipment Table

Functional dependencies:

- $\text{Equipment_ID} \rightarrow \text{Name}, \text{Model_Number}, \text{Manufacturer}, \text{Status}$

BCNF Verification:

- Only one functional dependency
- Determinant (Equipment_ID) is a superkey (primary key)

Status: BCNF COMPLIANT

2. Client table:

Functional dependencies:

- $\text{Client_ID} \rightarrow \text{First_Name}, \text{Last_Name}, \text{Contact_Email}, \text{Phone_Number}, \text{Organization}$
- $\text{Contact_Email} \rightarrow \text{Client_ID}, \text{First_Name}, \text{Last_Name}, \text{Phone_Number}, \text{Organization}$
- $\text{Phone_Number} \rightarrow \text{Client_ID}, \text{First_Name}, \text{Last_Name}, \text{Contact_Email}, \text{Organization}$

BCNF Verification:

- All determinants (Client_ID , Contact_Email , Phone_Number) are superkeys
- Each can uniquely identify a row

Status: BCNF COMPLIANT

3. Technician Table

Functional dependencies:

- $\text{Technician_ID} \rightarrow \text{First_Name}, \text{Last_Name}, \text{Certification}, \text{Contact_Email}, \text{Phone_Number}$
- $\text{Contact_Email} \rightarrow \text{Technician_ID}, \text{First_Name}, \text{Last_Name}, \text{Certification}, \text{Phone_Number}$

- $\text{Phone_Number} \rightarrow \text{Technician_ID}, \text{First_Name}, \text{Last_Name}, \text{Certification}, \text{Contact_Email}$

BCNF Verification:

- All determinants are superkeys
- Multiple candidate keys properly defined

Status: BCNF COMPLIANT

4. Test Table

Functional dependencies:

- $\text{Test_ID} \rightarrow \text{Test_Name}, \text{Description}$

BCNF Verification:

- Simple primary key dependency
- Determinant is a superkey

Status: BCNF COMPLIANT

5. Lab Table

Functional dependencies:

- $\text{Lab_ID} \rightarrow \text{Lab_Name}, \text{Location}, \text{Capacity}$

BCNF Verification:

- Primary key determines all attributes
- No other functional dependencies

Status: BCNF COMPLIANT

6. Booking Table

Functional dependencies:

- $\text{Booking_ID} \rightarrow \text{Booking_Date}, \text{Client_ID}, \text{Lab_ID}, \text{Start_Time}, \text{End_Time}, \text{Status}$

BCNF Verification:

- Only primary key dependency
- Determinant is a superkey

Status: BCNF COMPLIANT

7. Test_Session Table

Functional dependencies:

- Session_ID $\rightarrow$ Test_Date, Result, Duration, Technician_ID, Test_ID

BCNF Verification:

- Primary key determines all attributes
- No other functional dependencies

Status: BCNF COMPLIANT

8. Report Table

Functional dependencies:

- Report_ID $\rightarrow$ Report_Date, Findings, Approved_By, Session_ID
- Session_ID $\rightarrow$ Report_ID, Report_Date, Findings, Approved_By

BCNF Verification:

- Both determinants (Report_ID, Session_ID) are superkeys
- Each can uniquely identify a row

Status: BCNF COMPLIANT

9. Invoice Table

Functional dependencies:

- Invoice_ID $\rightarrow$ Booking_ID, Invoice_Date, Amount, Payment_Status
- Booking_ID $\rightarrow$ Invoice_ID, Invoice_Date, Amount, Payment_Status

BCNF Verification:

- Both determinants are superkeys (1:1 relationship)
- Each booking has exactly one invoice

Status: BCNF COMPLIANT

10. Maintenance_Record Table

Functional dependencies:

- Maintenance_ID $\rightarrow$ Equipment_ID, Maintenance_Date, Technician_ID, Notes

BCNF Verification:

- Primary key determines all attributes
- No other functional dependencies

Status: BCNF COMPLIANT

11. Booking_Equipment Table (Junction)

Functional dependencies:

- (Booking_ID, Equipment_ID) $\rightarrow$ (no additional attributes)

BCNF Verification:

- Only composite primary key
- No non-key attributes to create dependencies

Status: BCNF COMPLIANT

12. TestSession_Equipment Table (Junction)

Functional dependencies:

- (Session_ID, Equipment_ID) $\rightarrow$ (no additional attributes)

BCNF Verification:

- Only composite primary key
- No non-key attributes

Status: BCNF COMPLIANT

Table Name	All Determinants are Superkeys	BCNF Status
Equipment	Yes	COMPLIANT
Client	Yes	COMPLIANT
Technician	Yes	COMPLIANT
Test	Yes	COMPLIANT
Lab	Yes	COMPLIANT
Booking	Yes	COMPLIANT
Test_Session	Yes	COMPLIANT
Report	Yes	COMPLIANT
Invoice	Yes	COMPLIANT
Maintenance_Record	Yes	COMPLIANT
Booking_Equipment	Yes	COMPLIANT
TestSession_Equipment	Yes	COMPLIANT